21CSC529T Inferential Statistics

Session - 6

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Agenda

Unit 3: Estimation & Hypothesis Testing

- Resampling
- Sampling Distribution & Central limit theorem,
- Point estimate
- confidence interval
- Hypothesis Testing Basics
- Null & Alternative Hypothesis formulation - Types of hypotheses
- Hypothesis testing process
- Errors in hypothesis testing - Type I and Type II
- Power of test
- Number of tails
- Choice of test statistic

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Population and Sample

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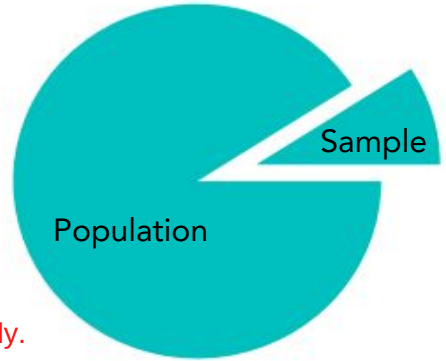
Population & Sample

- Population is a collection of all the individuals or objects

- Sample is a subset of the population which is a representative of the population



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Need to draw a sample

- In some of the real case scenarios we may not be able to gather all the observations in the population
- In such cases we can consider a sample which can be used to draw inferences about the population
- Sampling can reduce the computational time and cost
- In machine learning models, a sample obtained using an appropriate sampling strategy can provide accurate results with the less number of observations

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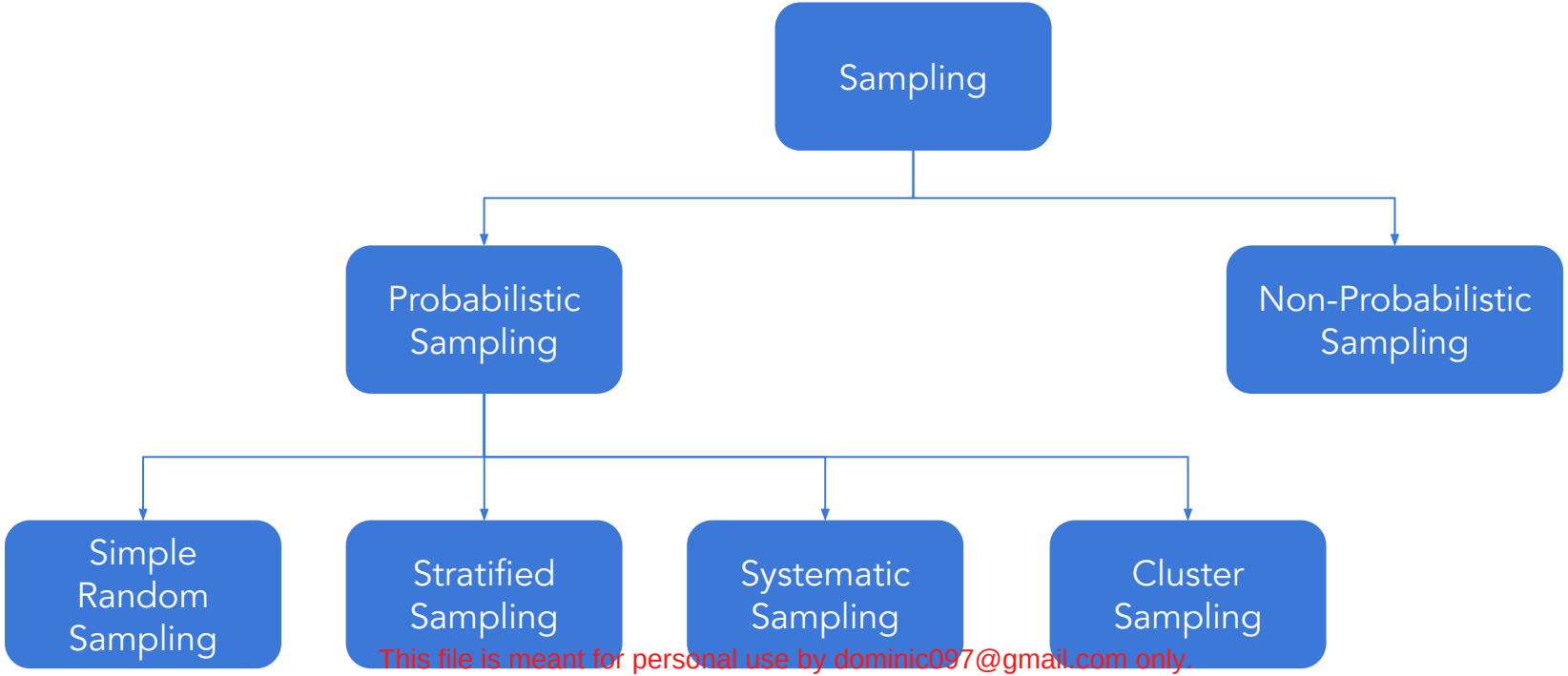
Sampling techniques



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Simple random sampling (SRS)

- Simplest and most common method
- Sample is drawn unit by unit
- Each unit has equal probability of being selected
- Two ways of selecting units:
  - Without replacement: if the selected unit is not returned to the population
  - With replacement: if the selected unit is returned to the population

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SRS without replacement

- At any specific stage in the sampling, the probability of any of the remaining units is 1/N, where N is the total number of sample units

- From N sample units, if a sample of size n is drawn, there are ${}^{N}C_{n}$ possible samples

- The probability of selecting one of them is $1/{}^{N}C_{n}$

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SRS without replacement

The unit selected to be in the sample is not returned to the population.

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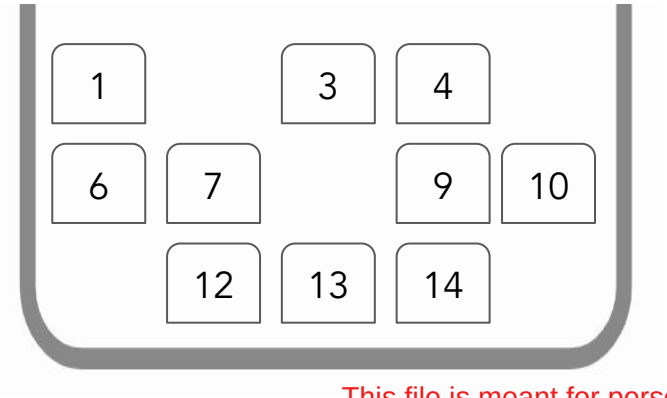
SRS without replacement - example

Drawing a sample from manufactured articles to know the proportion of defectives.

Manufactured articles



Sample to test for defective articles

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SRS with replacement

- The probability of selecting a specific unit at any given draw is always 1/N, where N is total number of sample units
- From N sample units, if sample of size n is drawn, there are N^n possible samples
- The probability of selecting one of them is $1/N^n$
- Example: You have 5 fruits and want to take a sample of 2. You can choose a fruit more than once.

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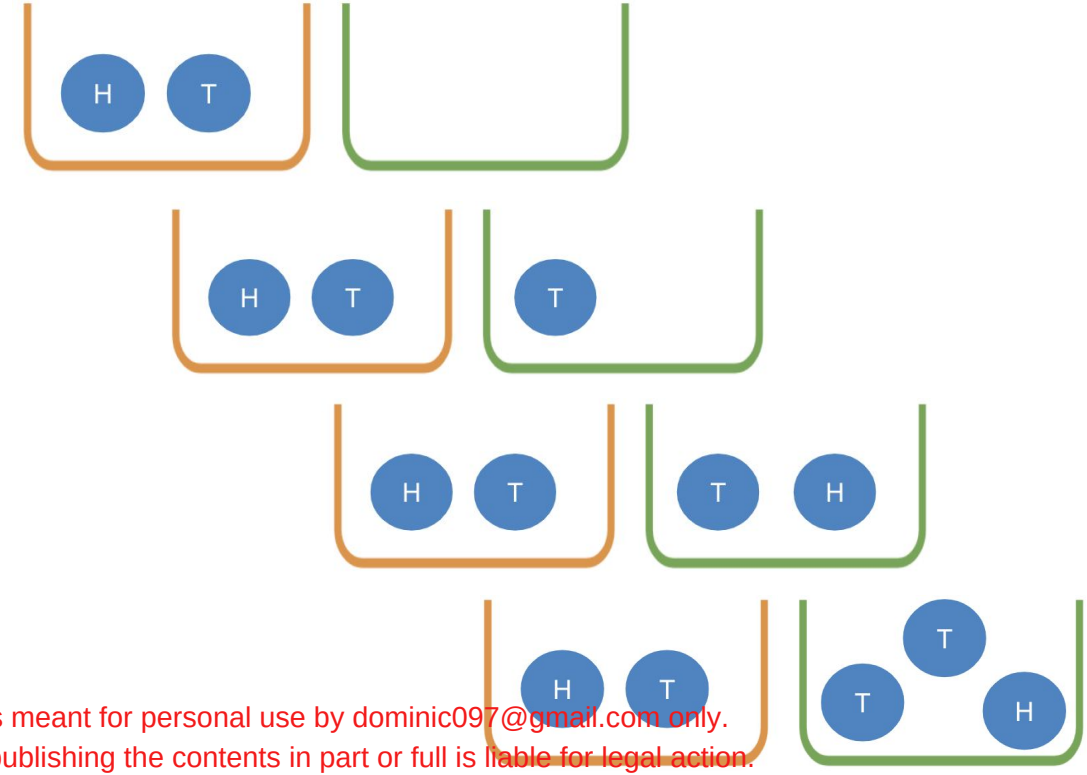
SRS with replacement



The unit selected to be in the sample is returned to the population.

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SRS with replacement - example

- Generally used in simulation studies
- A car insurance company wants to set the premium amount. The probabilities for major and minor accident are known to be 0.02% and 2% respectively
  (assume that an insuree can meet with an accident only once)
- A SRS with replacement is conducted to simulate 1 million such scenarios

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Summary

Sampling Method	Summary
Simple Random	Mostly used method. The sample can be selected with or without replacement.
Stratified	Sample point is selected from each subgroup in the population.
Systematic	Easy to implement for large population. The first sample point is selected randomly and remaining points are selected at fixed intervals.
Cluster	An entire subgroup in the population is selected as a sample point.

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Summary

- Sample is the subset of the population
- Different sampling methods: SRS with replacement (SRSWR), SRS without replacement (SRSWOR), Stratified, Systematic, Cluster sampling
- SRSWR method can be used if one wish to select the sample observation more than once
- Stratified sampling can be used when data is heterogeneous
- Cluster sampling can be used when the data consists of several groups

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Resampling

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Resampling

- Resampling is a technique in which we draw a sample from the available sample
- Resampling often involves performing similar testing methods with sample sizes within that group.
- This can mean testing the same sample, or reselecting samples that can provide more information about a population.
- The resampling techniques are:
  - Bootstrap resampling
  - Jackknife resampling

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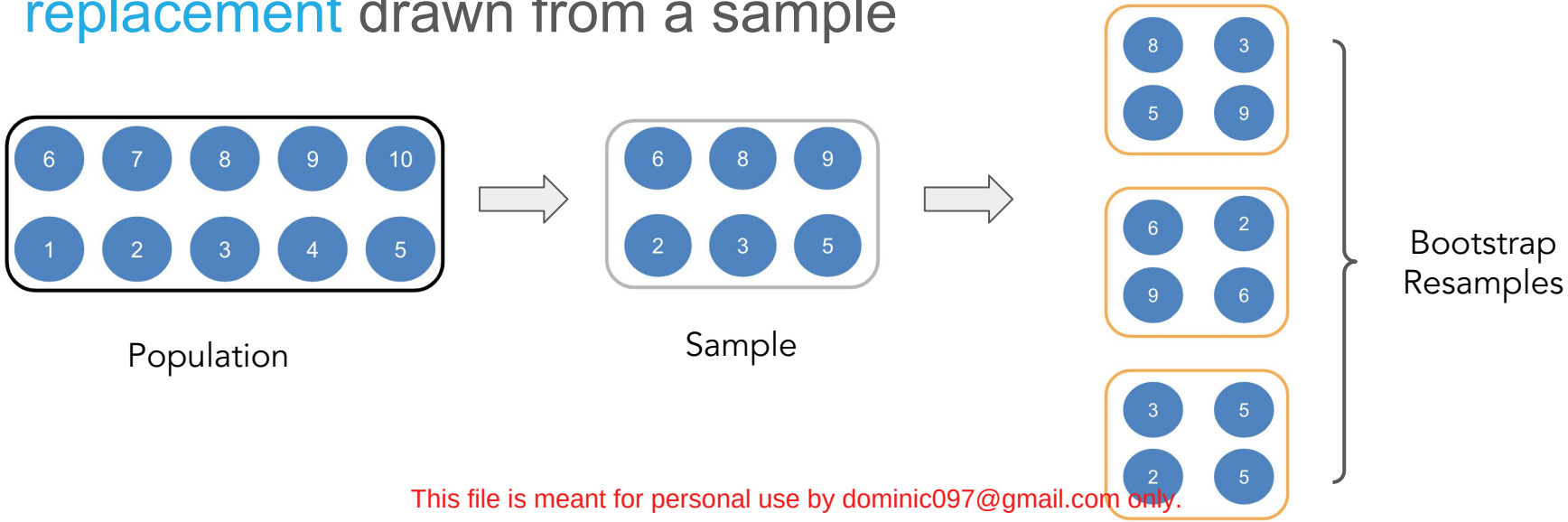
Bootstrap resample

Bootstrap resample is a simple random sample with replacement drawn from a sample



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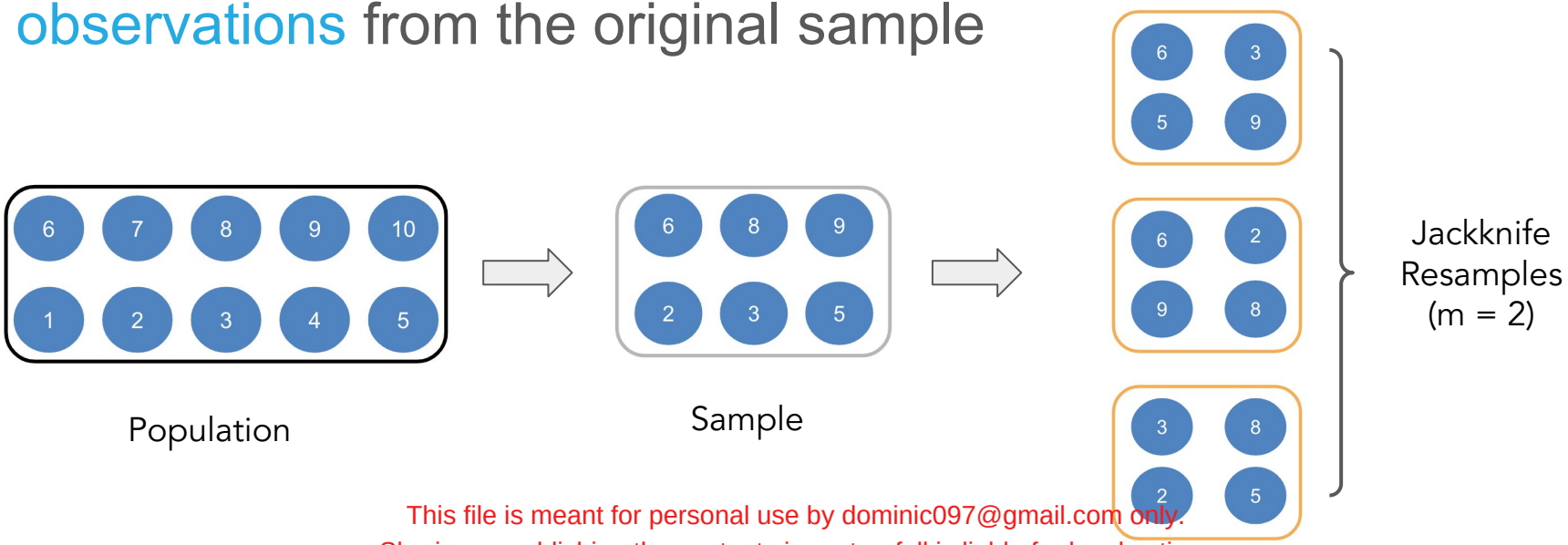
Jackknife resample

Jackknife resample is a method in which one deletes m observations from the original sample



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PLEASE!! NOTE

A machine is used for packing grains in a sack. The weight of every bag is recorded as follows:

15.6	15.7	14.6	14.7	15.8	15.0	15.1	14.9
14.8	15.5	15.4	15.3	14.8	15.1	14.8	14.9

Draw at least the two samples and obtain their means. Compare this with the population mean.

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Sampling variability

- Let us draw four different samples from the data
- The sample mean varies from sample to sample
- This intuitive idea is called sampling variability
- The sample mean is close to the population mean

Population Mean
15.125

Sample	Sample Mean
14.7, 15.1, 15.0, 15.6, 14.9, 15.4, 15.8, 15.1	15.20
15.8, 15.5, 15.3, 15.4, 15.8, 14.8, 14.6, 14.8	15.250
15.7, 15.6, 14.8, 14.8, 14.7,14.9, 15.6, 15.1	15.150
14.8, 15.1, 15.7, 14.6, 15.6, 15.4, 15.6, 15.0	15.225

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Parameter and Statistic

- Parameter:
  - The population quantities are termed as a parameter
  - In inferential statistics, sample is used to estimate the population parameters
  - Example: Population mean

- Statistic:
  - Statistical measure computed from sample observations
  - Also known as an estimator
  - Example: Sample mean

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Parameter and Statistic

The statistical notations for population parameters and sample statistics

	Population Parameter	Sample Statistic
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard Deviation	σ	s
Proportion	p	$\bar{p}$

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Sampling Distributions

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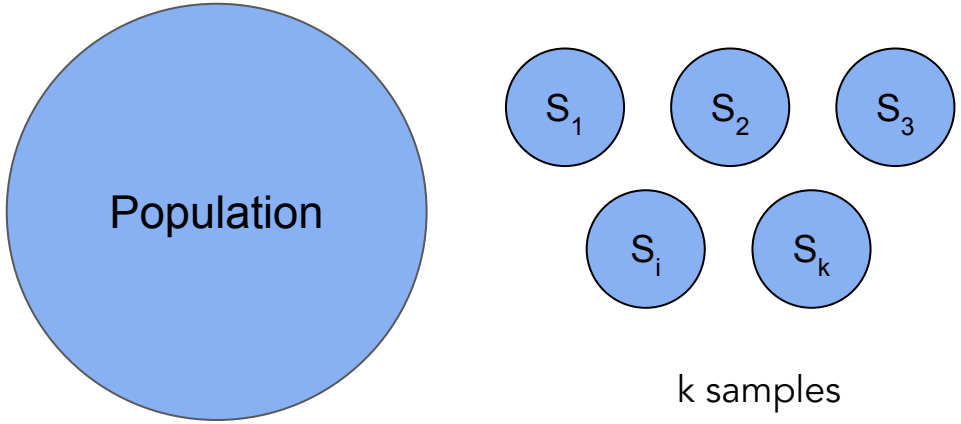
Sampling distribution

- A total of ${}^{N}C_{n}$ (= k say) samples can be drawn from a population of size N and sample size n

- Obtain the mean and variance of each of these samples



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Sampling distribution of sample mean

- The set of random sample means $\bar{x}_1$, $\bar{x}_2$,..., $\bar{x}_k$ are the observations for sampling distribution of sample mean

- These sample means follow the normal distribution

- Also, the mean of sample means is the population mean $\mu_{\bar{x}} = \mu$

Sample No	Sample Mean
1	$\bar{x}_1$
2	$\bar{x}_2$
3	$\bar{x}_3$
⋮	⋮
k	$\bar{x}_k$

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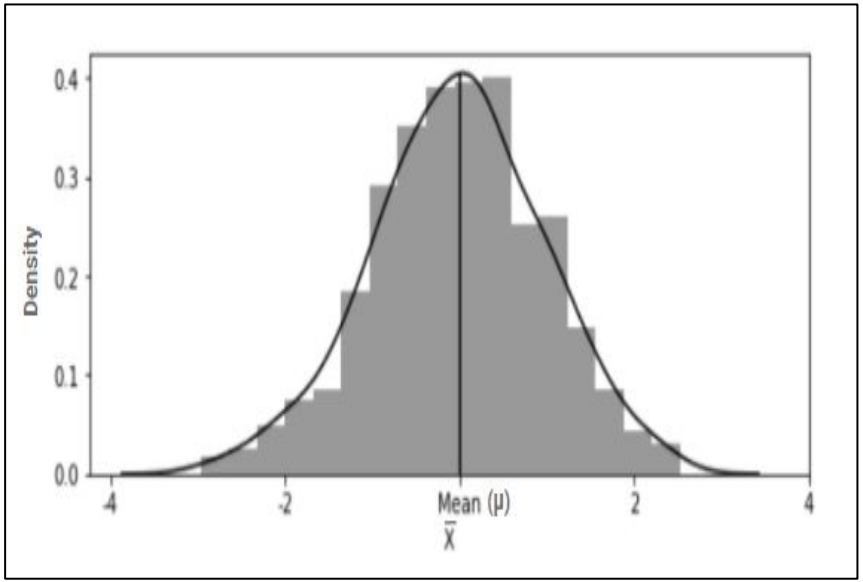
Sampling distribution of sample mean

- The mean of sampling distribution of sample mean is $\mu_{\bar{x}}$



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Central limit theorem

- The central limit theorem says that the sampling distribution of the mean will always be normally distributed, as long as the sample size is large enough.

- Regardless of whether the population has a normal, Poisson, binomial, or any other distribution, the sampling distribution of the mean will be normal.

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Central limit theorem

The central limit states that for random samples of large size n (usually n > 30) taken from a single population with mean μ and standard deviation σ, the sampling distribution of mean follows a normal distribution with mean $\mu_{\bar{x}}$ and standard deviation σ/√n or s/√n which ever is known.

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0, 1)$$

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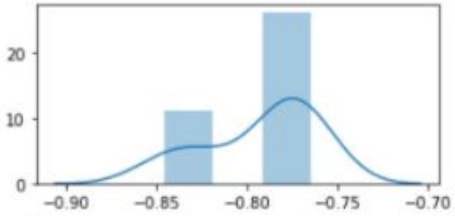
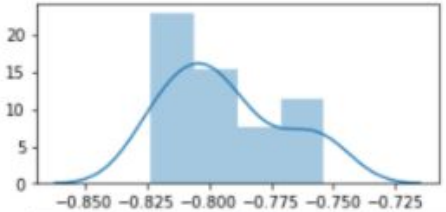
PLEASE!! NOTE

Central limit theorem

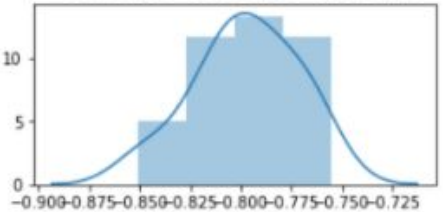
nSamples: 10



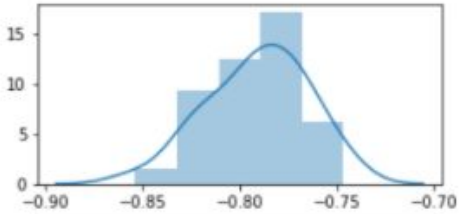
nSamples: 15



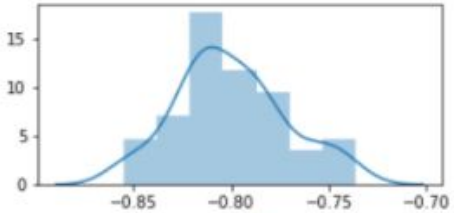
nSamples: 25



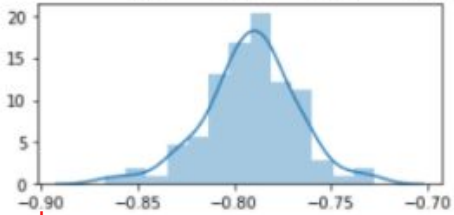
nSamples: 30



nSamples: 50



nSamples: 100



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A coffee shop claims that the average time customers spend in the shop is 10 minutes, with a standard deviation of 2 minutes. You take a random sample of 50 customers. What is the probability that the average time spent by these customers is greater than 10.5 minutes?

Solution:

Population mean (μ) = 10 minutes

Population standard deviation (σ) = 2 minutes

Sample size (n) = 50 customers

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{2}{\sqrt{50}} \approx \frac{2}{7.07} \approx 0.283$$

$$z = \frac{X - \mu}{SE} = \frac{10.5 - 10}{0.283} \approx \frac{0.5}{0.283} \approx 1.77$$

$$P(Z < 1.77) \approx 0.9616$$

To find $P(Z > 1.77)$:

$$P(Z > 1.77) = 1 - P(Z < 1.77) = 1 - 0.9616 \approx 0.0384$$

Conclusion:

The probability that the average time spent by the 50 customers is greater than 10.5 minutes is approximately 0.0384, or 3.84%.

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Inferential Statistics

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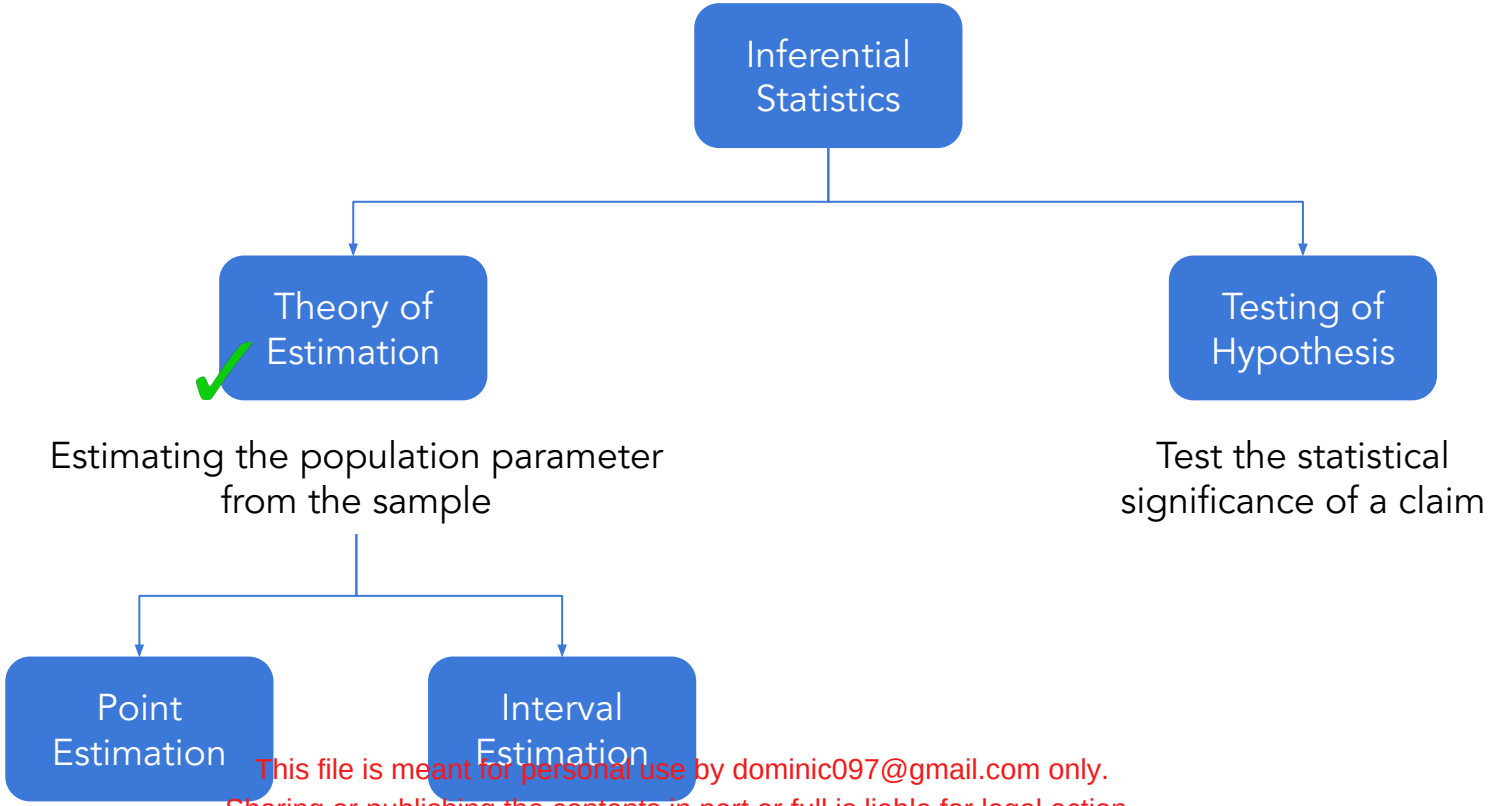
Inferential statistics



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Point Estimation

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Estimate

Question:

What would be the average income of a 27-year-old in London? How would you calculate it?

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Estimate

Solution:

The trivial way is to collect the income data of every 27-year-old, further compute its mean.

This mean would be the population mean.

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Point estimation

Is this scenario feasible? Is it possible to collect data of every 27-year-old person?

- To do so, draw a sample of size n and compute the mean.
- This sample mean is the point estimate of the population mean

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Point estimation

- The sample estimate is considered to be the point estimate of the population parameter
- If population parameters are not known they are estimated from the sample
- The point estimate of
  - population mean (μ) is sample mean ($\bar{X}$)
  - population variance (σ^2) is sample variance s^2
  - Population proportion (P) is sample proportion (p)

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Point Estimate

Question:

A financial firm has created 50 portfolios. From them a sample of 13 portfolios was selected, out of which 8 were found to be underperforming. Estimate the number of underforming portfolios?

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Point Estimate

Solution:

A financial firm has created 50 portfolios. Thus N = 50

From them a sample of 13 portfolios was selected. Thus n = 13

The desired category is of underperforming portfolios. 8 of 13 were found to be underperforming. Thus x = 8

To find: Estimate the number of underperforming portfolios

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Point Estimate

Solution:

The estimate of proportion of underperforming portfolios is given by

$$\hat{p} = \frac{x}{n} = \frac{8}{13}$$

To estimate the number of underperforming portfolios multiply by the estimate by N.

$$\text{The number of estimated underperforming samples} = \hat{p}.\,N = \frac{8}{13}.\,50 = 30.76 \approx 31$$

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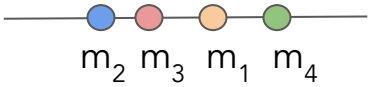
Demerits of point estimate

- In most of the scenarios, the point estimate is not accurate
- It is the single value representation of the population that differs with each sample
- The deviation of the point estimate from the population parameter gives rise to sampling error

Sample means of different samples



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Sampling error

- Sampling error is the absolute difference between the population parameter and its sample statistic (point estimate)

- Since the entire population is not considered as the sample, the values of mean, median, quantiles, and so on calculated on sample differ from the actual population values

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Sample size

Question:

The average temperature on a summer morning in Alaska is 55°F. The research student of climate studies records the temperature for 15 summer mornings in ^{o}F.

Data = [49.8, 52.3, 51, 56.9, 54.8, 63, 58.2, 54.1, 50.4, 49.2, 47, 51.3, 43.5, 56, 55]

Find the sampling error for the average temperature.

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Interval Estimation

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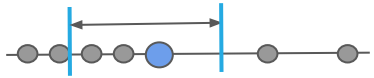
Need of interval estimation

- The point estimate is taken from a particular sample
- For a different sample, we get a different estimate (sampling variability)
- This problem is resolved if we obtained the interval estimate

Point estimate



Interval estimate

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Interval Estimation

- Interval estimate is a range or an interval within which the parameter lies for a stated confidence level

- Based on the central limit theorem

- It is known as the confidence interval

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Confidence level

- Confidence level (1-α) is the percentage of all possible point estimates that can be expected to include the actual population parameter
- α is known as the level of significance
- The higher confidence level illustrates the wider confidence interval
- A 95% confidence level implies that 95% of the confidence intervals would include the actual population parameter

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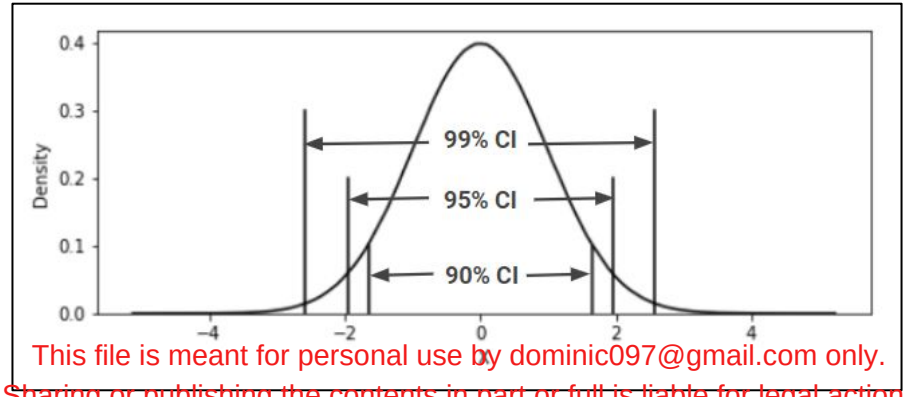
Confidence level

Example:

The average income estimate for a 27-year-old in London is £45,000. After taking repeated samples, the 99% confidence intervals of the average income include the actual population parameter for 99% of the times.



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Recall - CLT

For random samples of large size n taken from a single population with mean μ and standard deviation σ, the sampling distribution of mean follows a normal distribution with mean $\mu_{\bar{x}}$ and standard deviation $\sigma/\sqrt{n}$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

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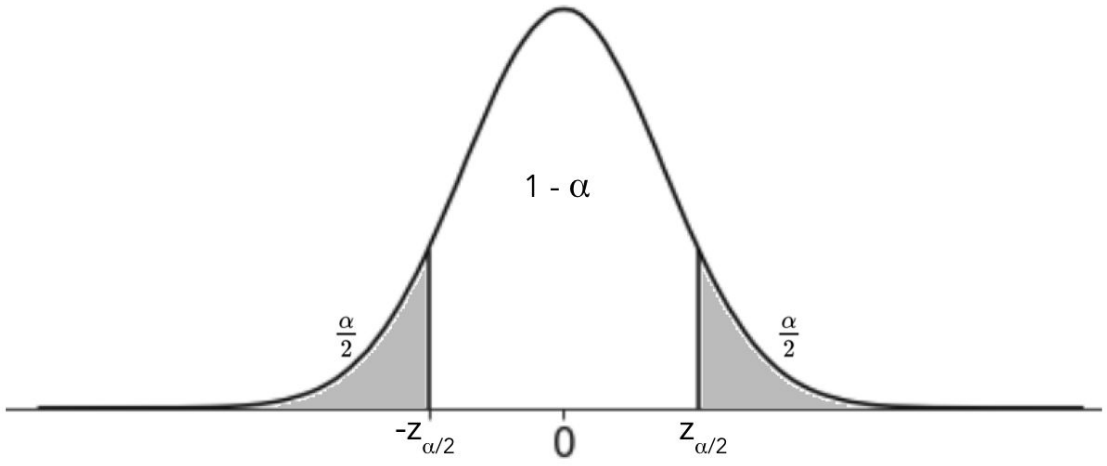
Interval estimate

From the CLT, $P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$



Where α is is the level of significance

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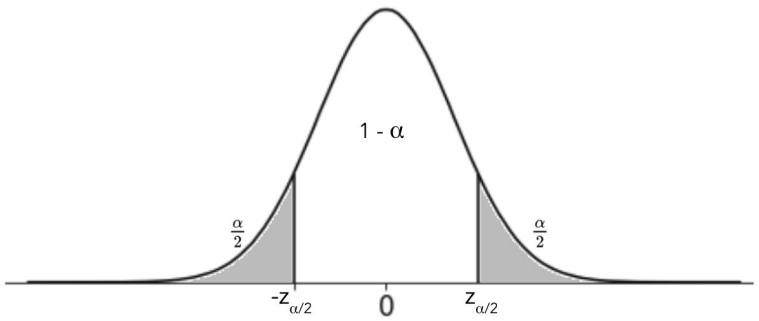
Interval estimate

From the CLT, $P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$

Value of α	$Z_{\alpha/2}$	Value of $Z_{\alpha/2}$
0.10	$z_{0.05}$	1.645
0.05	$z_{0.025}$	1.96
0.01	$z_{0.005}$	2.575



Note: To obtain these values in python use scipy.stats.norm.isf()

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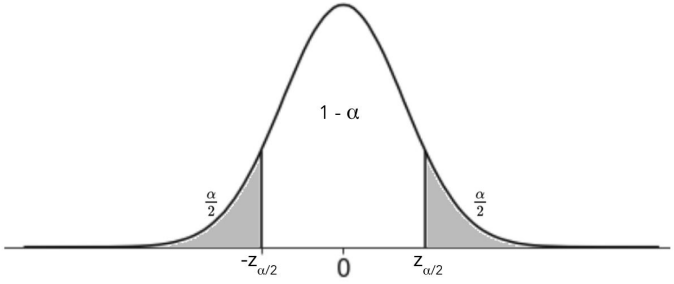
Interval estimate

From the CLT, $P(-z_{\alpha/2} \le Z \le z_{\alpha/2}) = 1 - \alpha$



and value of Z is $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$

Thus, $P(-z_{\alpha/2} \le \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \le z_{\alpha/2}) = 1 - \alpha$

$$\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

Rearranging the terms, $P(\bar{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}) = 1 - \alpha$

This defines $(1 - \alpha)\%$ confidence interval

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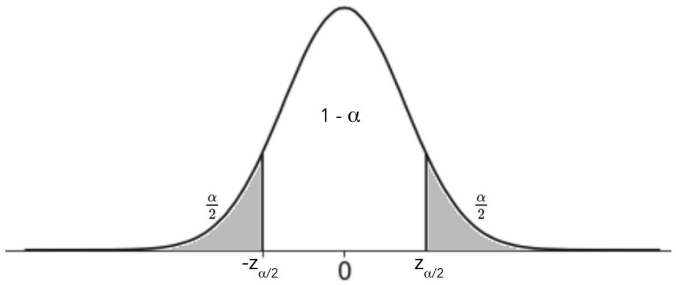
Interval estimate

The 100(1 - α)% confidence interval is

$$\underbrace{\bar{X}}_{\text{Point Estimate}} \pm \underbrace{z_{\alpha/2}\frac{\sigma}{\sqrt{n}}}_{\text{Margin of error}}$$



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Margin of error

Question:

100 bags of coal were tested and had an average of 35% of ash with a standard deviation of 15%. Calculate the margin of error for a 90% confidence level.

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Margin of error

Solution:

100 bags of coal were tested and had an average of 35% of ash with a standard deviation of 15%.

Here $n = 100$

$\sigma = 0.15$

To find: The margin of error for a 90% confidence level.

Here $\alpha = 0.10$. Thus $z_{\alpha/2} = 1.64$

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Margin of error

Solution:

The margin of error for a 90% confidence level is

$$z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 1.64\frac{0.15}{\sqrt{100}} = 0.0246$$

A 90% CI with 2.46% margin of error implies that the sample mean will be within 2.46% points of the real population value 90% of the time

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PLEASE!! NOTE

- As sample size increases the margin of error decreases

- If margin of error is fixed then the sample size can be obtained

$$E = z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

$$\Rightarrow n = z^2_{\alpha/2}\left(\frac{\sigma^2}{E^2}\right)$$

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PLEASE!! NOTE

Sample size

Question:

For the previous question, for a margin of error 2.46. Verify whether you get sample size of 100.

Previous Question: 100 bags of coal were tested and had an average of 35% of ash with a standard deviation of 15%. Calculate the margin of error for a 90% confidence level.

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Interval estimate

Question:

From a sample of 250 observations it is found that average income of a 27-year-old Londoner is £45,000 with a sample standard deviation of £4000. Obtain the 95% confidence interval to estimate the average income.

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Interval estimate

Solution:

A sample of 250 observations of a 27-year-old Londoner has an average income of £45,000 with a sample standard deviation of £4000

Thus, $\bar{X}$ = 45,000, σ = 4000 and n = 250

To find: The 95% confidence interval.

Here α = 0.05. So $z_{0.025}$ = 1.96

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Interval estimate

Solution:

The 95% confidence interval is

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$\Rightarrow 45000 \pm 1.96 \frac{4000}{\sqrt{250}}$$

$$\Rightarrow (44504.155, 45495.845)$$

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PLEASE!! NOTE

- It cannot be said that $P(44504.155 \leq \mu \leq 45495.845) = 0.95$

- The population parameter μ is a constant

- It can be said that from many samples drawn of size 250, construct a 95% CI for each sample, roughly 95% of the intervals would actually contain the population parameter μ

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Confidence Interval

Question:

For the previous question, obtain the 90% and 99% confidence intervals. Further, logically explain which one of them is the widest.

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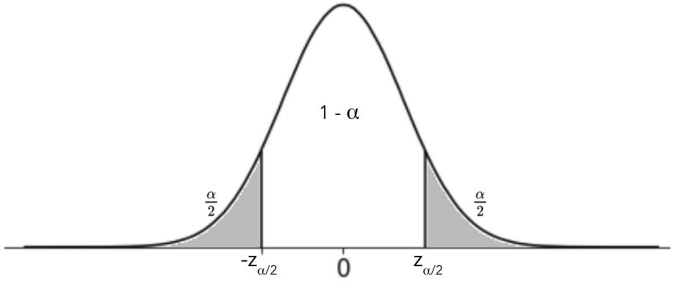
Interval estimate

The 100(1 - α)% confidence interval for proportion is

$$\underbrace{\hat{p}}_{\text{Point Estimate}} \pm \underbrace{z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}_{\text{Margin of error}}$$



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Interval estimate

Question:

A financial firm has created 50 portfolios. From them a sample of 13 portfolios was selected, out of which 8 were found to be underperforming. Construct a 99% confidence interval to estimate the population proportion.

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Interval estimate

Solution:

From them a sample of 13 portfolios was selected. Thus n = 13

The desired category is of underperforming portfolios. 8 of 13 were found to be underperforming. Thus x = 8

Thus the estimate of sample proportion is x/n = 8/13 =0.615

To find: Construct a 99% confidence interval to estimate the population proportion.

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Interval estimate

Solution:

Here α = 0.01. So $z_{0.005}$ = 2.575

The 99% confidence interval is

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\Rightarrow 0.615 \pm 2.575\sqrt{\frac{0.615(1-0.615)}{13}}$$

= (0.23, 0.96)

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A manufacturer's rechargeable batteries (sample of n=100) have mean life $\bar{x}=1002$ charges. Sample standard deviation is known to be $\sigma=14$. Find a **95%** CI for the true mean.

- Point estimate: $\bar{x}=1002$
- Standard error: $SE=\sigma/\text{sqrt}(n) = 1.4$.
- 95% z critical value: 1.96.
- Margin of error: $MoE=z_{0.975}\times SE=1.96\times 1.4=2.744$.
- CI: $\bar{x}\pm MoE=1002\pm 2.744=[999.256,\ 1004.744]$.

Interpretation: We are 95% confident the true mean battery life lies between **999.26** and **1004.74** charges.

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Summary

- Sample estimate of a population parameter is considered as a point estimate
- Sample mean, median, standard deviation are the examples of the point estimate
- Point estimate varies significantly for different samples
- Confidence interval returns the interval estimate for the population parameter
- The confidence level describes the uncertainty of the estimate
- Confidence interval is calculated using the point estimate and the margin of error
- The optimal sample size can be obtained by deciding the required margin of error

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Testing of Hypothesis

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Question:

A manufacturer produces batteries. He claims that the life of his batteries is 25,500 hours. Can we verify his claim?

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Solution:

A trivial solution is to obtain the life of all the batteries produced by the manufacturer by using them and thereafter verify this claim

However, by doing so all the manufactured products (batteries) will be exhausted and he will not have any batteries to sell.

To avoid such scenarios, we statistically test the hypothesis

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What is a hypothesis?

A hypothesis is a claim about the population parameter

Examples:

- The average diameter of bolts manufactured is 200 cm
- The new drug is more effective

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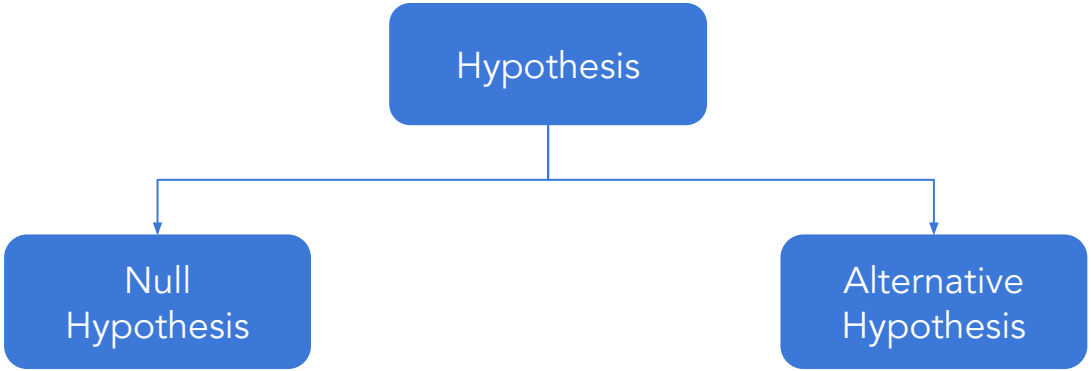
Hypothesis - example

A pharmaceutical has claimed to have produced a new drug is effective against cancer treatment.

To test the pharmaceuticals claim, first write the test hypotheses -
Null Hypothesis and the Alternative Hypothesis



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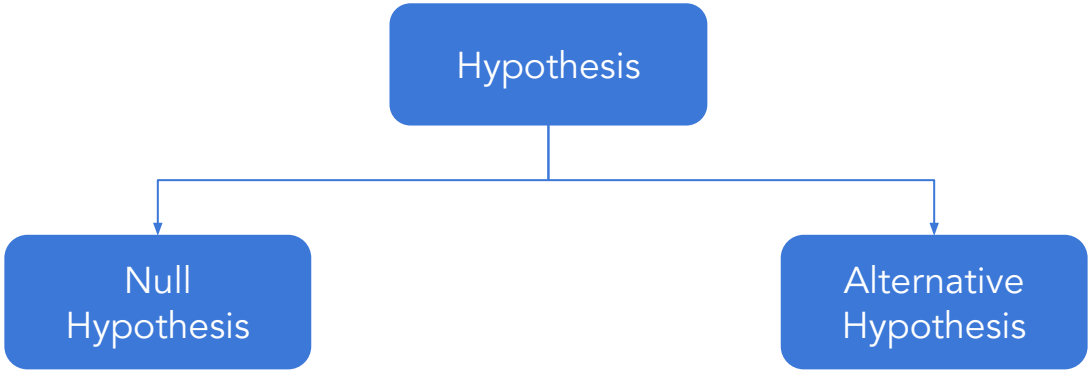
Null and alternative hypothesis



- Hypothesis of no change
- Denoted by H_0

- It directly contradicts the null hypothesis
- Also known as research hypothesis
- Denoted by H_a or H_1

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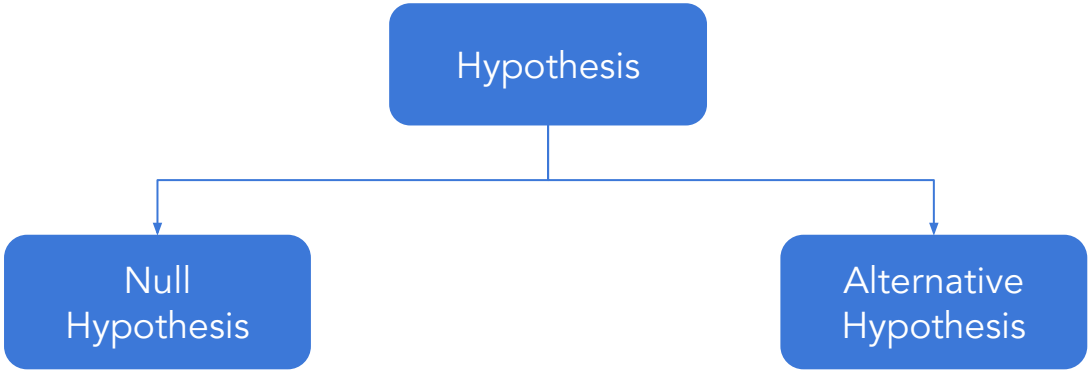
Hypothesis - example

A pharmaceutical has claimed to have produced a new drug which treats the patients against cancer.



H_0: The new drug is not effective

H_1: The new drug is effective

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Hypothesis - example

- Now consider the effectiveness of the new drug is determined by the number of days a patient requires to recover.
- The data for the number of days the patient take to recover are recorded
- Let the random variable X be the number of days required to recover
- Thus μ is the average number of days required to recover

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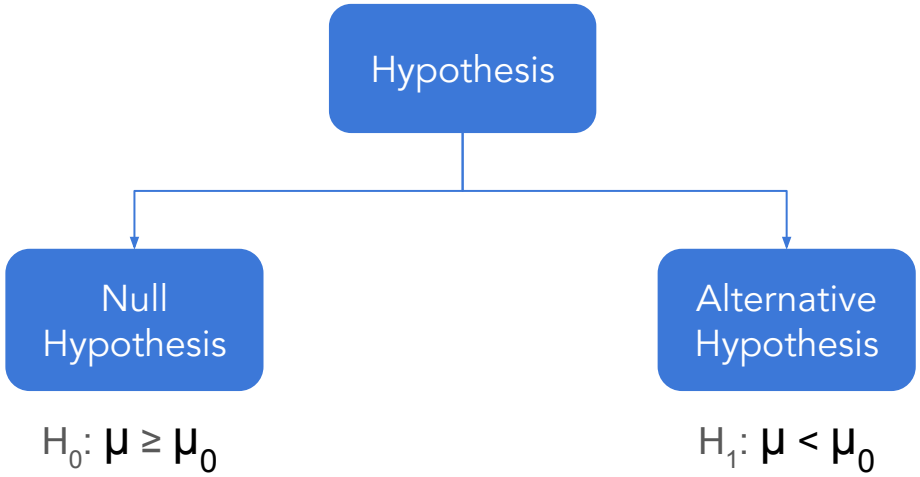
State the hypothesis

It is claimed that the new drug reduces the recovery time

μ: the average number of days required to recover

Here, μ_0 is the general recovery time



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Hypothesis testing

- The null and alternative hypothesis are contrasting
- Either the null or the alternative hypothesis can be true
- The decision of the test procedure is one of the following:
  - Reject H_0 implies the null hypothesis is false
  - Fail to reject H_0 implies the null hypothesis is true

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PLEASE!! NOTE

"Fail to reject H_0"

- The null hypothesis is assumed to be true unless there is sufficient evidence to reject it.(we have insufficient evidence to support H1)
- Failing to reject H0 does not confirm that H0 is true; it only means there is not enough evidence against it.
- The alternative hypothesis has the burden of proof, i.e. from the data it will have enough evidence to say the null hypothesis is false
- Hence, it is preferred to say 'fail to reject H_0' instead of accept H_0

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Errors in Hypothesis Testing

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Probability of committing an error

- P(Type I error) is the probability of wrongly rejecting a true null hypothesis
- It is the level of significance (α) set for the hypothesis
- An α of 0.05 suggests that one is willing to accept a 5% chance that you reject a true null hypothesis
- In order to lower this risk, set a lower α. However, it will be more difficult to reject the false null hypothesis

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Error in hypothesis testing

		Test Decision	
		Reject H_0	Accept H_0
Actual Situation	H_0 is True	Type I error Wrongly reject true H_0	Correct Conclusion Correctly accept true H_0
	H_0 is False	Correct Conclusion Correctly reject false H_0	Type II error Wrongly accept false H_0

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Probability of committing an error

- P(Type II error) is the probability of wrongly accepting a false null hypothesis
- Denoted by β
- 1 - β is the power of the test, i.e. correctly rejecting a false null hypothesis
- In order to lower the risk of type II error, ensure the power of the test is high
- In python, we calculate the power of a Z test using the 'power.zt_ind_solve_power()' from statsmodels library

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Controlling the errors

- Both errors cannot be controlled simultaneously
- Fix one error and minimize the other
- Generally α is fixed and β is controlled

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Previously

- The unknown population parameters are estimated by from sample
- The unknown population parameters are replaced by the sample estimates
- Test the hypothesis for population mean when the population variance is known

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Two-tailed, Left-tailed, Right-tailed Tests

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Left-tailed Test

$$H_0: \mu \geq 200$$

$$H_1: \mu < 200$$

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Left-tailed Test

$$H_0: \mu \geq 200$$

$$H_1: \mu < 200$$

Points Left

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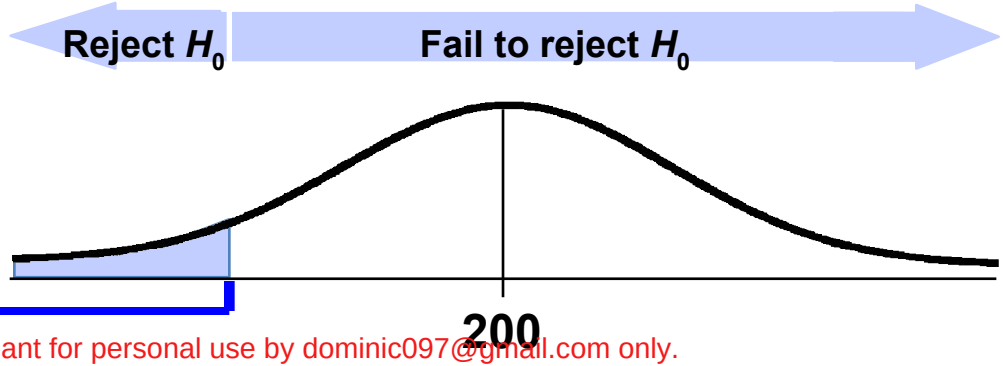
Left-tailed Test

$$H_0: \mu \geq 200$$

$$H_1: \mu < 200$$

Points Left



Values that differ significantly from 200

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Right-tailed Test

H_0: $\mu \leq 200$

H_1: $\mu > 200$

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Right-tailed Test

H_0: $\mu \leq 200$

H_1: $\mu > 200$

Points Right

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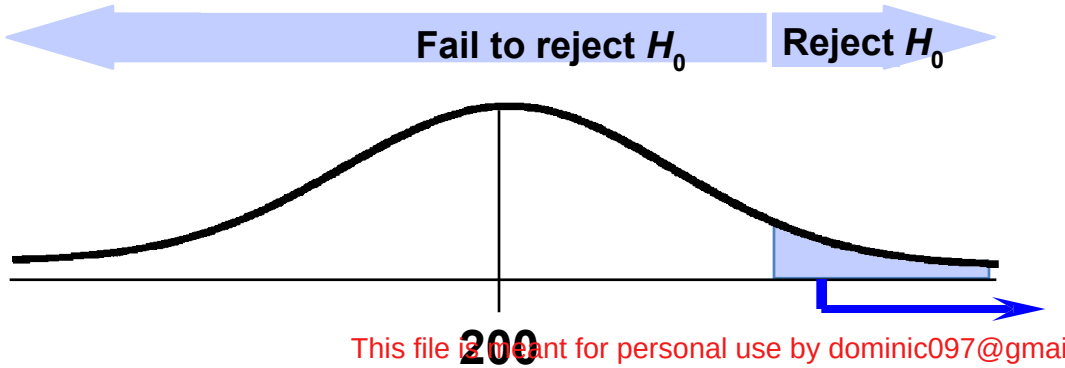
Right-tailed Test

H_0: μ ≤ 200

H_1: μ > 200

Points Right



Values that differ significantly from 200

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Two-tailed Test

$$H_0: \mu = 200$$

$$H_1: \mu \neq 200$$

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Two-tailed Test

$$H_0: \mu = 200$$

$$H_1: \mu \neq 200$$

α is divided equally between the two tails of the critical region

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Two-tailed Test

$$H_0: \mu = 200$$
$$H_1: \mu \neq 200$$

Means less than or greater than

α is divided equally between the two tails of the critical region

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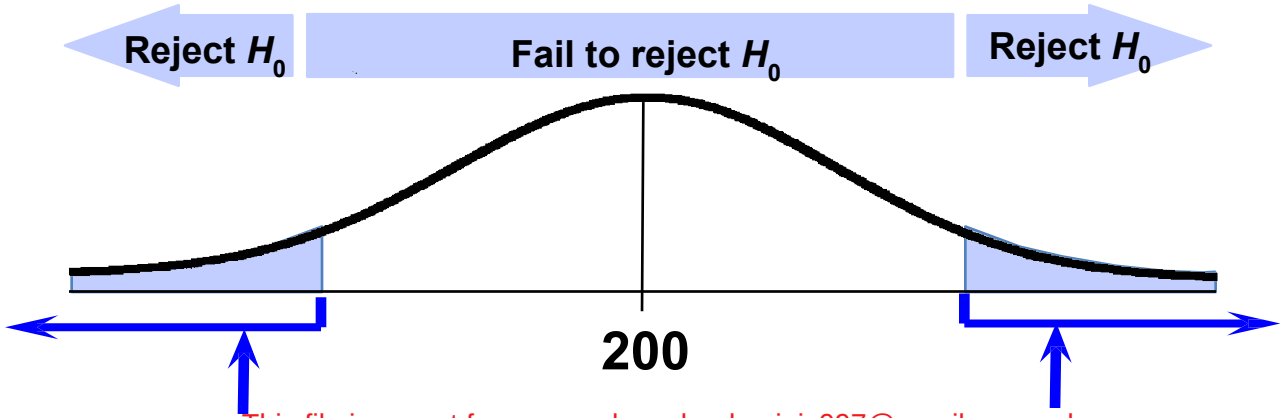
Two-tailed Test

$$H_0: \mu = 200$$
$$H_1: \mu \neq 200$$

Means less than or greater than

α is divided equally between the two tails of the critical region



Values that differ significantly from 200

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Question:

Write the null and alternative hypothesis for the following scenarios:

1. A bolt manufacturing company claims that the average number of bolt manufactures in a day is 60
2. The new virus testing kit takes 20 hours less time than usual testing kit that takes 34 hours for the results
3. An analyst wants to test whether the Apple Inc. has outperformed in 2019 by 13.5%

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Question:

Write the null and alternative hypothesis for the following scenarios:

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1. A bolt manufacturing company claims that the average number of bolts he manufactures in a day is 60

Solution:

The null and alternative hypothesis for the following scenarios:

Let X: the number of bolts manufactured in a day and

μ: the average number of bolts manufactured in a day

$H_0: \mu = 60$ Against $H_1: \mu \neq 60$

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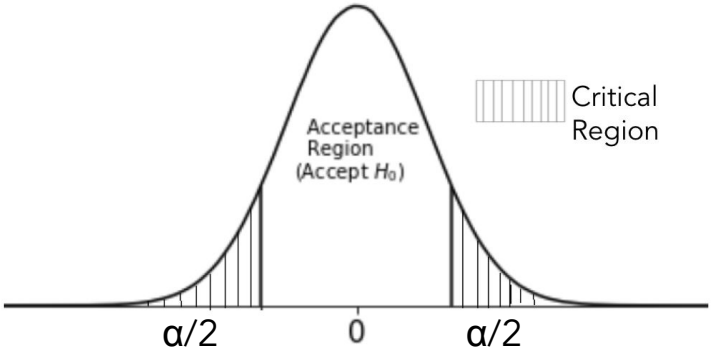
Two tailed test

Two-tailed test is a method in which the critical region of a distribution is split in its two tails. It tests whether a sample is greater than or less than the population parameter.

A critical region is a region in which the null hypothesis is rejected. It is also known as ‘Rejection Region’.



Example

Let X: the number of bolts manufactured in a day and

μ: the average number of bolts manufactured in a day

H_0: μ = 60 Against H_1: μ ≠ 60

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2. The new virus testing kit takes less time than usual testing kit that takes 34 hours for the results

Solution:

The null and alternative hypothesis for the following scenarios:

Let X: the time required for the results by a testing kit

μ: the average time required for the results by a testing kit

$H_0: \mu \geq 34$ Against $H_1: \mu < 34$

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Left tailed test

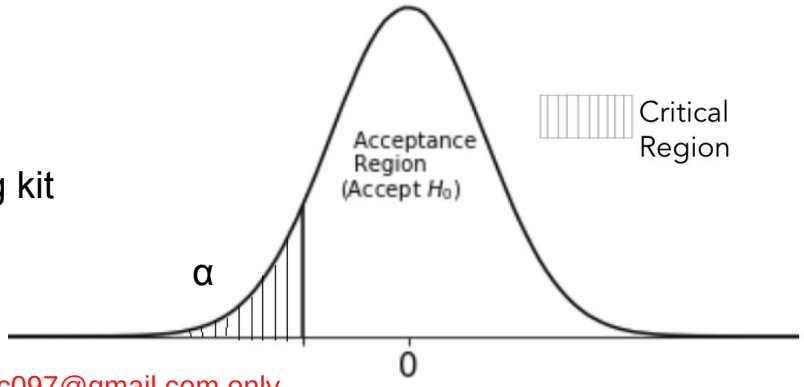
A left-tailed test is a statistical hypothesis test set up to show that the sample parameter would be lower than the population parameter.

Example

Let X: the time required for the results by a testing kit

μ: the average time required for the results by a testing kit

H_0: μ ≥ 34 Against H_1: μ < 34



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3. An analyst wants to test whether the Apple Inc. has outperformed in 2019 by 13.5%

Solution:

The null and alternative hypothesis for the following scenarios:

Let X: the performance of Apple Inc. stock

μ: the average performance of Apple Inc. stock

$H_0: \mu \leq 13.5$ Against $H_1: \mu > 13.5$

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Right tailed test

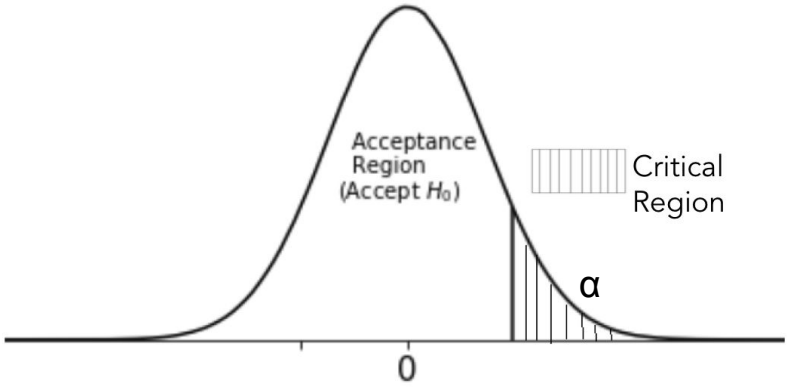
A right-tailed test is a statistical hypothesis test set up to show that the sample parameter would be higher than the population parameter.

Example

Let X: the performance of Apple Inc. stock

μ: the average performance of Apple Inc. stock

H_0: $\mu \leq 13.5$ Against H_1: $\mu > 13.5$



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Summary

Example	Hypothesis	Test Type
The company manager claims that the average weight of cookies in a gift box is 235 gm	$H_0: \mu = 235$ $H_1: \mu \neq 235$	Two Tailed Test
The station officer claims that the bullet train takes less than 2 hours from London to Brussels	$H_0: \mu \geq 2$ $H_1: \mu < 2$	Left Tailed Test
To test whether the average number of boat rides are more than 10 per day	$H_0: \mu \leq 10$ $H_1: \mu > 10$	Right Tailed Test

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Write the null and alternative hypothesis and identify the type of the test.

- To test the hypothesis that the mean diastolic blood pressure for a group of 85 adults is less than 90mm
- The tensile strength of an alloy is more than 500 Pa
- The average heart rate of a healthy human is 85 beats per minute
- The yield BT cotton is more than 450 kg lint per hectare

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Seven Steps of Hypothesis Testing

1) Describe in words the population characteristic about which hypotheses are to be tested

2) State the null hypothesis, H_o

3) State the alternative hypothesis, H_1 or H_a

4) Display the test statistic to be used

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Seven Steps…

5) Identify the rejection region

- Is it an upper, lower, or two-tailed test?
- Determine the critical value associated with α, the level of significance of the test

6) Compute all the quantities in the test statistic, and compute the test statistic itself

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Seven Steps…

7) State the conclusion.

- Decide whether to reject the null hypothesis, H_o, or fail to reject the null hypothesis.
- Conclusion depends on the level of significance of the test.
- Remember to state your result in the context of the specific problem.

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A local telephone company claims that the average length of a phone call is 8 minutes. In a random sample of 58 phone calls, the sample mean was 7.8 minutes and the standard deviation was 0.5 minutes. Is there enough evidence to support this claim at $\alpha = 0.05$?

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Two-Tailed Test - Example

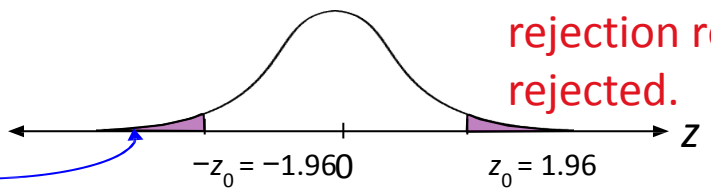
Example

A local telephone company claims that the average length of a phone call is 8 minutes. In a random sample of 58 phone calls, the sample mean was 7.8 minutes and the standard deviation was 0.5 minutes. Is there enough evidence to support this claim at $\alpha = 0.05$?

$H_0: \mu = 8$ (Claim) $H_a: \mu \neq 8$

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{7.8 - 8}{0.5 / \sqrt{58}}$$

$$\approx -3.05.$$



The test statistic falls in the rejection region, so H_0 is rejected.

At the 5% level of significance, there is enough evidence to reject the claim that the average length of a phone call is 8 minutes.

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A manufacturer claims that its rechargeable batteries are good for an average of more than 1,000 charges. A random sample of 100 batteries has a mean life of 1002 charges and a standard deviation of 14. Is there enough evidence to support this claim at $\alpha = 0.01$?

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One – Tailed Test Example

Example:

A manufacturer claims that its rechargeable batteries are good for an average of more than 1,000 charges. A random sample of 100 batteries has a mean life of 1002 charges and a standard deviation of 14. Is there enough evidence to support this claim at $\alpha = 0.01$?

$H_0: \mu \leq 1000$ $H_a: \mu > 1000$ (Claim)

The level of significance is $\alpha = 0.01$.

The standardized test statistic is

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{1002 - 1000}{14/\sqrt{100}}$$

$$= 1.43$$

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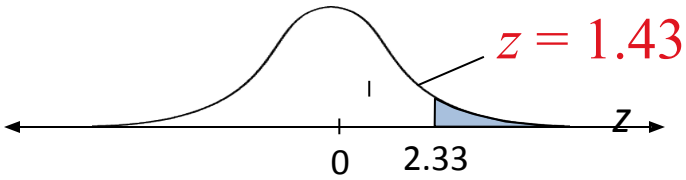
One-Tailed Test – Example Contd…

Example continued:

A manufacturer claims that its rechargeable batteries are good for an average of more than 1,000 charges. A random sample of 100 batteries has a mean life of 1002 charges and a standard deviation of 14. Is there enough evidence to support this claim at $\alpha = 0.01$?

$H_0: \mu \leq 1000$ $H_a: \mu > 1000$ (Claim)

$z = 1.43$



Z *statistic* is lesser than $\alpha = 0.01$, fail to reject H_0.

At the 1% level of significance, there is not enough evidence to support the claim that the rechargeable battery has an average life of more than 1000 charges.

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A health organization claims that the average cholesterol level for adult men is 200 mg/dL. To verify this claim, a researcher takes a random sample of 100 adult men and finds that the average cholesterol level in this sample is 205 mg/dL, with a population standard deviation of 30 mg/dL.

We want to test at a significance level of 0.05 to determine if there is enough evidence to reject the health organization's claim.

- Null Hypothesis (H_0): The average cholesterol level is 200 mg/dL. $\mu=200$
- Alternative Hypothesis (H_a): The average cholesterol level is not 200 mg/dL (two-tailed). $M\neq200$

$$z = \frac{x - \mu}{\sigma / \sqrt{n}} = \frac{205 - 200}{30/\sqrt{100}}$$

$$\approx 1.67$$

The critical Z-values are approximately ± 1.96

Since 1.67 falls between -1.96 and 1.96, we fail to reject the null hypothesis.

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A company claims that 80% of its customers are satisfied with its product. To test this claim, a random sample of 100 customers is surveyed, and 70 customers report that they are satisfied. Do the data provide sufficient evidence to reject the company's claim at the 0.05 significance level?

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Soln:

Null Hypothesis (H0): p≥0.80 (80% of customers are satisfied)
Alternative Hypothesis (Ha): p<0.80 (fewer than 80% of customers are satisfied)

Sample Size (n) = 100
Number of satisfied customers (X) = 70
Sample Proportion p^ =X/n = 70/100 = 0.70

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Where:

- $\hat{p}$ = sample proportion
- p_0 = hypothesized population proportion (0.80)
- n = sample size

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$$Z = \frac{0.70 - 0.80}{\sqrt{\frac{0.80(1-0.80)}{100}}} = \frac{-0.10}{\sqrt{\frac{0.80 \times 0.20}{100}}} = \frac{-0.10}{\sqrt{0.16/100}} = \frac{-0.10}{0.04} = -2.5$$

For a one-tailed test at α=0.05, the critical z-value can be found using the Z-table is approximately -1.645.

Now, we compare the calculated Z-value with the critical Z-value:

Calculated Z=−2.5
Critical Z=−1.645
Since −2.5<−1.645, we reject the null hypothesis.

Conclusion

There is sufficient evidence to reject the company's claim that 80% of its customers are satisfied. The data suggests that fewer than 80% of customers are satisfied at the 0.05 significance level.

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